Real-Time Campus Aeroallergen Dispersion Modeling Using Gaussian Plume Theory and Satellite-Based Vegetation Detection

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Abstract

We present Campus AeroAllergen Mapping (CAM), a web-based system for modeling hyperlocal pollen dispersion across a school campus at meter-scale resolution. The framework couples a two-dimensional potential-flow wind field, which deflects and channels the wind around building footprints, with a steady-state Gaussian plume model that disperses pollen along the local flow, evaluating concentration at student breathing height ($z = 1.5$ m). Individual tree canopies are identified from high-resolution satellite imagery using color-based segmentation with watershed splitting, and building rooftops are detected with the Segment Anything Model (SAM); both are refined through human-in-the-loop correction. Temporal emission variability is captured by species-specific Gaussian phenological gate functions, and building footprints define indoor exclusion zones where concentration is zeroed. The system is applied to two campuses—Henry M. Gunn High School (Palo Alto, CA) and the Stanford University core—modeling thousands of individual trees and buildings, and produces concentration fields at meter scale in a few seconds. A fixed annual color scale enables intuitive seasonal comparison, showing summer concentrations at roughly 6% of the spring peak. The system integrates exposure along student walking paths to report lower-exposure routes; these predictions have not yet been validated against measured pollen counts.

Keywords: potential-flow wind field, Gaussian plume model, pollen dispersion, aeroallergen mapping, satellite vegetation detection, deep-learning segmentation, phenological modeling

1 Introduction

Allergic rhinitis affects 10–30% of the global population, with airborne pollen constituting the dominant environmental trigger [Bousquet et al., 2008]. On school campuses where students transit between buildings through corridors lined with allergenic flora, localized pollen concentrations can vary by orders of magnitude over distances of tens of meters due to source proximity, wind channeling, and structural aerodynamic effects.

Existing pollen monitoring relies on regional stations reporting area-averaged counts at temporal resolutions of hours to days [Sofiev et al., 2013]. These measurements cannot resolve the hyperlocal concentration gradients determining individual exposure during a 5-minute walking transit. We address this gap by developing a physics-based framework that:

1. Computes a 2D potential-flow wind field that deflects and channels the wind around campus buildings;

2. Solves the atmospheric advection-diffusion equation analytically (Gaussian plume) at meter-scale resolution, driven by the local wind at each source;
3. Identifies pollen sources (tree canopies) and building rooftops from satellite imagery, the latter with a deep-learning segmentation model;
4. Modulates emissions temporally using species-specific phenological functions;
5. Masks indoor building volumes from the exposure field.

Among these, the building-resolving wind field is the central methodological contribution of this work. A conventional plume model assumes a single, spatially uniform wind over the whole domain; on an open field this is adequate, but a school campus is a *micro-environment* in which clusters of buildings continuously divert, channel, and locally accelerate the wind that carries pollen between them. Resolving that building-scale flow is what turns a generic plume model into a micro-environment model, and it is the piece we treat first (Section 2) and quantify with direct simulation (Section 2.4). The system is deployed on two campuses of contrasting scale and layout: Henry M. Gunn High School (Palo Alto, CA), a 25-hectare campus, and the core of Stanford University; together they contain thousands of individually mapped trees and buildings.

2 Modeling the Campus Micro-Environment: A 2D Potential-Flow Wind Field

2.1 Why a Uniform Wind Is Not Enough

The defining feature of a campus is that it is densely built. Between a lecture hall and a gymnasium the wind does not blow in a straight line: it stagnates against windward walls, accelerates around corners, and is squeezed—channeled—into the gaps and corridors that students actually walk through. Because pollen is carried by the wind, these building-scale flow distortions directly reshape where pollen goes. A pollen model that assumes one uniform wind vector for the entire campus cannot, even in principle, represent a plume bending around a building to reach a courtyard on the far side, or pooling in a sheltered, low-wind alley. Capturing this is the motivation for computing an explicit, spatially varying wind field before any pollen is released, and using the *local* wind at each tree to drive that tree's plume.

2.2 Governing Equations

We model the horizontal, near-surface wind as an incompressible, irrotational (*potential*) flow. Incompressibility means the flow neither piles up nor thins out ($\nabla \cdot \mathbf{u} = 0$); irrotationality means it carries no ambient spin, so the velocity is the gradient of a single scalar velocity potential ϕ:

$$\mathbf{u} = (u, v) = \nabla \phi = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right). \tag{1}$$

Combining these two statements yields **Laplace's equation** for the potential:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0. \tag{2}$$

The physics enters entirely through the boundary conditions:

- **Free stream (domain edge, Dirichlet).** Far from the buildings the wind equals the ambient value reported for the campus. We impose $\phi = u_\infty x + v_\infty y$ on the outer boundary, where (u_∞, v_∞) are the components of the ambient wind (converted from meteorological speed and direction).

- **No penetration on walls (Neumann).** Wind cannot flow through a building. On every building face the velocity component normal to the wall is zero, $\partial\phi/\partial n = 0$. This single condition is what forces the flow to go *around* each footprint rather than through it, and hence produces all of the diversion, stagnation, and channeling behavior.

2.3 Numerical Solution

Equation (2) is solved on the same 120×120 regular grid used later for the concentration field, covering the full campus ($640\,\text{m} \times 560\,\text{m}$, cell size ≈ 5 m). Building footprints—detected from satellite imagery and refined by a reviewer (Section 5.5)—are rasterized onto this grid to mark "solid" cells; every other cell is "fluid."

We relax the discrete Laplacian to convergence with **red-black Gauss-Seidel successive over-relaxation (SOR)**. Each fluid cell is updated toward the average of its four neighbors; where a neighbor is a solid (building) cell, that neighbor is replaced by the cell's own value, which enforces the zero-normal-gradient (no-penetration) wall condition exactly. The grid cells are two-colored like a checkerboard and the two colors are updated in sequence, which makes each sweep a true Gauss-Seidel step for which over-relaxation is numerically stable; the over-relaxation factor is set to its theoretical optimum $\omega = 2/(1 + \sin(\pi/N))$ for an N-cell grid, giving rapid convergence (a full campus solve completes in about one second). Once ϕ has converged, the velocity field follows from Eq. (1) by finite differences, and cells inside buildings are set to zero velocity.

2.4 Simulated Wind Flow Around Buildings

Figure 1 contrasts the conventional uniform-wind assumption with the computed potential-flow field for an idealized cluster of buildings under a 4 m/s westerly wind. The uniform assumption (panel a) is the same vector everywhere. The potential-flow field (panel b) instead shows the wind stagnating (low speed, blue) against the windward faces, accelerating around the corners (high speed, red), and channeling through the gap between the two building rows—a direct, physically grounded picture of how the built environment reorganizes the wind. Figure 2 shows the same computation on the real Henry M. Gunn High School footprints for two ambient wind directions (southwesterly and westerly). The streamlines bend around building clusters and speed up in the narrow passages between them, and the pattern shifts with wind direction, illustrating that the wind field must be recomputed whenever the ambient conditions change.

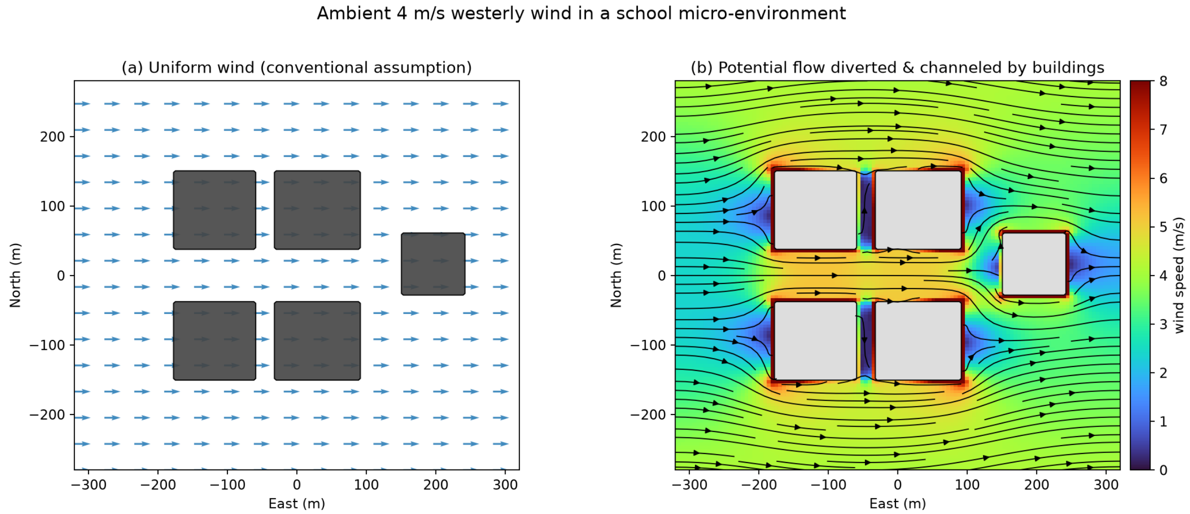

Figure 1: Wind in a school micro-environment under a 4 m/s westerly ambient wind. (a) The conventional uniform-wind assumption: an identical vector everywhere. (b) The computed 2D potential-flow field (background: wind speed; black curves: streamlines). The wind stagnates against windward faces (blue), accelerates around corners (red), and is channeled through the gaps between buildings—structure that a uniform wind cannot represent. The color scale is clipped at twice the free-stream speed; potential flow is singular at sharp convex corners, so the very highest corner speeds are grid-limited.

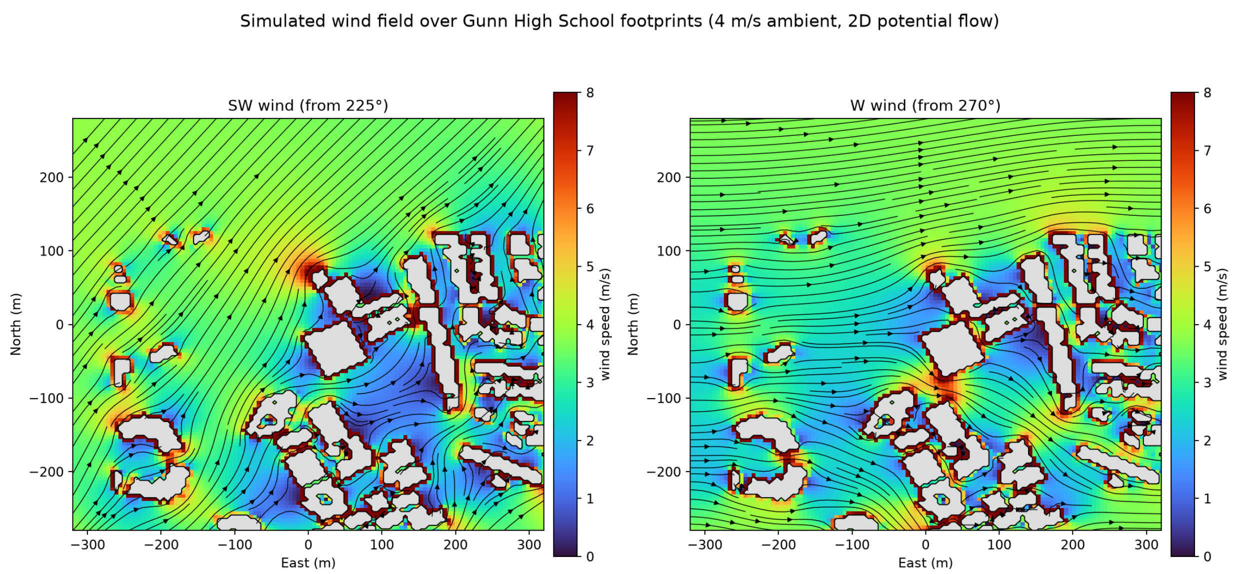

Figure 2: Simulated potential-flow wind field over the real building footprints of Henry M. Gunn High School (4 m/s ambient), for a southwesterly wind (left) and a westerly wind (right). Streamlines (black) divert around building clusters while the flow accelerates (red) in the narrow passages between them and slows (blue) in sheltered zones; the local wind at each tree drives that tree's pollen plume (Section 3.7).

Honest limitation. Potential flow is inviscid: it has no boundary-layer separation, and therefore no turbulent wake or recirculation zone behind a building. It captures how wind is *diverted*, channeled, and accelerated by obstacles, but not the low-velocity, pollen-trapping lee zone immediately downwind of a building, which would require a viscous (e.g. RANS/CFD) treatment beyond the scope of a real-time interactive system. It is used here as a computationally cheap, physically motivated approximation to the building-scale wind variation—which is precisely the effect a uniform wind omits entirely.

3 Atmospheric Dispersion Model

3.1 Physical Picture

When a tree releases pollen grains into the air, two physical processes determine where those grains end up:

1. **Advection** (bulk transport by wind): The wind carries the pollen cloud bodily in the downwind direction, like a river carrying a dye patch. If the wind blows from the southwest at 3 m/s, the pollen cloud moves northeast at 3 m/s.
2. **Diffusion** (turbulent spreading): Atmospheric turbulence—random gusts and eddies—causes the pollen cloud to spread out in all directions as it travels. The cloud starts concentrated near the tree and becomes progressively wider and more dilute downwind, forming a cone-shaped plume.

The combination produces a characteristic pattern: highest concentration immediately downwind of the tree, decreasing with distance as the plume spreads, and zero concentration upwind.

3.2 Governing Equation

These processes are described mathematically by the *advection-diffusion equation*:

$$\underbrace{\frac{\partial C}{\partial t}}_{\text{concentration change}} + \underbrace{u\frac{\partial C}{\partial x} + v\frac{\partial C}{\partial y}}_{\text{wind carries pollen (advection)}} = \underbrace{K_x\frac{\partial^2 C}{\partial x^2} + K_y\frac{\partial^2 C}{\partial y^2} + K_z\frac{\partial^2 C}{\partial z^2}}_{\text{turbulence spreads pollen (diffusion)}} - \underbrace{\lambda C}_{\text{settling/removal}} + \underbrace{Q\delta(\mathbf{x}-\mathbf{x}_s)}_{\text{tree emitting pollen}} \tag{3}$$

Each term has a clear physical meaning:

- $C(x,y,z,t)$ — **Pollen concentration** (grains/m^3) at position (x,y,z) and time t. This is what a student breathes in.
- (u,v) — **Wind velocity components** (m/s). u is the east-west wind speed, v is the north-south wind speed. Measured from weather station or API. A 4 m/s southwest wind means $u \approx 2.8$ m/s (eastward) and $v \approx 2.8$ m/s (northward).
- K_x, K_y, K_z — **Turbulent eddy diffusivity** (m^2/s). These quantify how vigorously the atmosphere mixes. On a calm, stable morning ($K \approx 1$ m^2/s), pollen stays concentrated near the source. On a windy, sunny afternoon ($K \approx 50$ m^2/s), turbulent eddies rapidly dilute the pollen cloud. K_z (vertical) is typically smaller than K_x, K_y (horizontal) because vertical mixing is suppressed by gravity.

- λ — **Deposition/removal rate** (s^{-1}). Pollen is continuously removed from the air by gravitational settling onto the ground, impaction on surfaces, and washout by rain. Heavier pollen (Pine, 60 μm diameter) has λ about 5$\times$ larger than light pollen (Oak, 20 μm), meaning Pine falls out of the air much faster.
- Q — **Source emission rate** (grains/s). How many pollen grains the tree releases per second. A mature Oak at peak bloom emits ~500 grains/s from its canopy.
- $\delta(\mathbf{x} - \mathbf{x}_s)$ — **Dirac delta function**. Indicates that emission occurs at a single point $\mathbf{x}_s$ (the tree's location). This is the "point source" approximation.

3.3 Gaussian Plume Solution

Solving the full PDE numerically at every timestep would be computationally expensive. Instead, we use the *steady-state* analytical solution (assuming wind conditions change slowly relative to pollen transport time). This yields the famous **Gaussian plume** formula [Pasquill, 1961]:

$$C(x', y', z) = \underbrace{\frac{Q}{2\pi U \sigma_y \sigma_z}}_{\text{dilution factor}} \cdot \underbrace{\exp\left(-\frac{y'^2}{2\sigma_y^2}\right)}_{\text{crosswind spread}} \cdot \underbrace{V(z)}_{\text{vertical profile}} \tag{4}$$

Each factor has a physical interpretation:

- $\dfrac{Q}{2\pi U \sigma_y \sigma_z}$ — **Dilution factor**. The faster the wind (U large) and the wider the plume (σ_y, σ_z large), the more diluted the pollen becomes. Doubling wind speed halves the concentration everywhere.
- $\exp(-y'^2/2\sigma_y^2)$ — **Crosswind Gaussian profile**. Concentration is highest on the plume centerline ($y' = 0$, directly downwind) and drops off as a bell curve to either side. σ_y is the "width" of this bell curve—how far sideways the pollen has spread.
- $V(z)$ — **Vertical profile with ground reflection**:

$$V(z) = \underbrace{\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right)}_{\text{direct path from source}} + \underbrace{\exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)}_{\text{ground-reflected path}} \tag{5}$$

The first term is pollen arriving directly from the tree canopy at height H. The second term accounts for pollen that has hit the ground and "bounced" back up (mathematically modeled as a mirror image source below ground). This ensures no pollen is lost through the ground surface.

3.4 Key Parameters Explained

3.4.1 σ_y and σ_z: Plume Spread (meters)

These are the most important parameters. They control how wide (σ_y) and tall (σ_z) the plume is at any given downwind distance:

- **Near the tree** ($x' = 10$ m): $\sigma_y \approx 0.8$ m, $\sigma_z \approx 0.6$ m. The plume is narrow—only a few meters wide. Concentration is very high.

- **50 m downwind**: $\sigma_y \approx 3.5$ m, $\sigma_z \approx 2.5$ m. The plume has spread to about 7 m wide.
- **200 m downwind**: $\sigma_y \approx 12$ m, $\sigma_z \approx 8$ m. The plume is now 24 m wide but much more dilute.

They grow with distance because turbulence has more time to mix the pollen sideways and vertically. The growth rate depends on *atmospheric stability*:

$$\sigma_y = a_y \cdot (x')^{b_y}, \quad \sigma_z = a_z \cdot (x')^{b_z} \tag{6}$$

3.4.2 Atmospheric Stability Classes (A–F)

Stability describes how much the atmosphere resists or encourages vertical mixing:

- **Class A (very unstable)**: Hot sunny afternoon. Strong updrafts vigorously mix pollen vertically. Plume spreads rapidly, diluting quickly. *Paradoxically, unstable conditions produce lower ground-level concentrations because pollen is lofted upward.*
- **Class D (neutral)**: Overcast, moderate wind. Default condition. Moderate mixing.
- **Class F (very stable)**: Clear, calm night or early morning. Very little vertical mixing. Plume stays narrow and concentrated near emission height. *Worst case for ground-level exposure if source is low.*

We estimate stability from two observable quantities:

- Wind speed < 2 m/s + clear sky $\rightarrow$ Class A/B (unstable, strong solar heating)
- Wind speed 3–5 m/s + partly cloudy $\rightarrow$ Class C/D (neutral)
- Wind speed > 6 m/s $\rightarrow$ Class D (mechanical turbulence dominates)

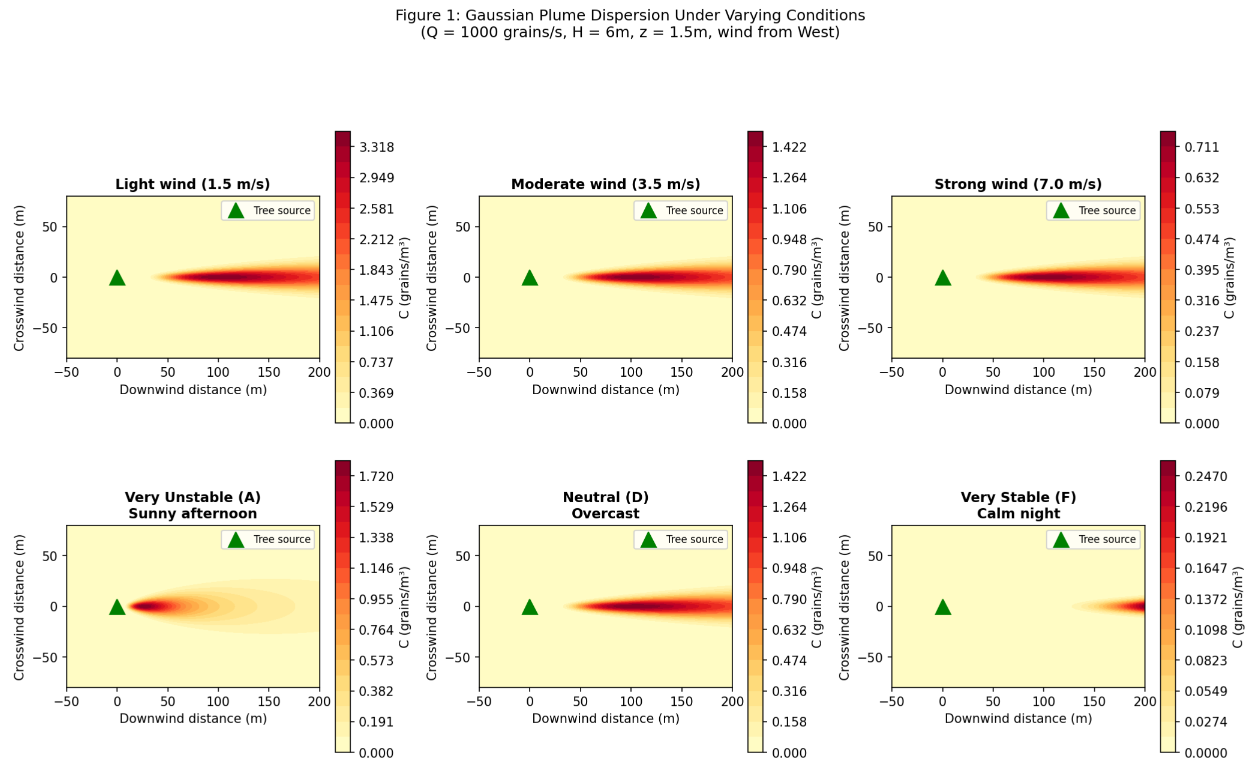

Figure 3: Gaussian plume concentration fields under varying conditions. Top row: effect of wind speed (stability class D). Stronger winds dilute and elongate the plume. Bottom row: effect of atmospheric stability (wind = 3.5 m/s). Unstable conditions (A) spread pollen rapidly; stable conditions (F) confine it in a narrow, concentrated plume.

3.4.3 U: Wind Speed (m/s)

Mean horizontal wind speed at reference height (typically 10 m). Obtained from OpenWeatherMap API. Physical effects:

- **Strong wind** ($U > 5$ m/s): Pollen is rapidly carried far downwind but heavily diluted. Concentration drops quickly with distance.
- **Light wind** ($U < 2$ m/s): Pollen lingers near the source tree. High concentrations in a small area. The model sets a minimum of $U = 0.5$ m/s to avoid division by zero (calm conditions are handled as very light drift).

3.4.4 H: Effective Source Height (meters)

The height at which pollen enters the atmosphere. For trees, this is the canopy height where catkins/flowers release grains. We use $H = 6.0$ m (average for campus trees ranging from 4 m shrubs to 15 m oaks). The concentration at ground level ($z = 1.5$ m) depends on how far pollen must fall/mix downward from height H.

3.4.5 z: Receptor Height (meters)

Fixed at $z = 1.5$ m—the height of a student's nose and mouth (breathing zone). This is where inhaled dose matters for allergic response.

3.4.6 (x', y'): Wind-Aligned Coordinates

The plume always points downwind. To compute concentration at any campus location, we transform from map coordinates to a frame aligned with the current wind direction:

$$x' = \Delta x \cos\alpha + \Delta y \sin\alpha \quad \text{(downwind distance from tree)} \tag{7}$$
$$y' = -\Delta x \sin\alpha + \Delta y \cos\alpha \quad \text{(crosswind offset)} \tag{8}$$

where $\alpha = \frac{\pi}{180}(270° - \theta)$ converts the meteorological wind direction θ (degrees clockwise from North) to a mathematical angle. Key consequences:

- If you are directly downwind of a tree ($y' = 0$): maximum exposure.
- If you are perpendicular to the wind ($x' = 0$): zero concentration (upwind or crosswind).
- If the wind shifts direction, the entire plume rotates to point the new way.

3.5 Species-Dependent Particle Dynamics

Pollen grains vary significantly in aerodynamic properties across species, affecting their atmospheric transport behavior. Three key species-dependent parameters modify the dispersion model:

3.5.1 Gravitational Settling Velocity

Pollen grains are not neutrally buoyant; they settle under gravity at a terminal velocity v_s governed by Stokes' law for spherical particles in laminar flow:

$$v_s = \frac{\rho_p \, d_p^2 \, g}{18 \, \mu} \tag{9}$$

where ρ_p is the pollen grain density ($\mathrm{kg\,m^{-3}}$), d_p is the aerodynamic diameter (m), $g = 9.81\ \mathrm{m\,s^{-2}}$ is gravitational acceleration, and $\mu = 1.81 \times 10^{-5}$ Pa·s is the dynamic viscosity of air at 20°C.

Table 1 lists the physical properties and resulting settling velocities for each campus species. Pine pollen, with its large diameter ($\sim$60 μm), settles at 3.1 cm/s—roughly 6× faster than the small Oak grains ($\sim$22 μm, 0.5 cm/s). This means Pine pollen deposits closer to the source tree, while Oak pollen remains airborne longer and disperses further downwind.

Table 1: Pollen grain physical properties and settling velocities.

Species	d_p (μm)	ρ_p (kg/m^3)	v_s (cm/s)	Airborne half-life
Valley Oak	22	1050	0.5	Long (hours)
Coast Live Oak	20	1050	0.4	Long (hours)
Pine	58	550	3.1	Short (minutes)
Redwood	30	800	1.0	Moderate
Cedar	45	560	1.5	Moderate
Chinese Elm	28	1000	1.1	Moderate
Sycamore	20	1020	0.4	Long (hours)

3.5.2 Tilted Plume: Gravitational Modification

The settling velocity tilts the plume centerline downward. The effective plume height decreases with downwind distance:

$$H_{eff}(x') = H - \frac{v_{s,i} \cdot x'}{U} \tag{10}$$

where H is the initial emission height, $v_{s,i}$ is the species-specific settling velocity, x' is the downwind distance, and U is the wind speed. For heavy Pine pollen in light winds ($U = 2$ m/s), the plume centerline drops by 1 m over just 65 m of travel, causing rapid ground-level concentration increase near the source.

Substituting into the Gaussian plume (Eq. 4), the species-dependent vertical term becomes:

$$V_i(z, x') = \exp\left(-\frac{(z - H_{eff,i})^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z + H_{eff,i})^2}{2\sigma_z^2}\right) \tag{11}$$

3.5.3 Deposition Velocity and First-Order Decay

Pollen removal from the atmosphere occurs through gravitational settling (dry deposition) and impaction on surfaces. The effective decay rate λ_i in Eq. (3) is species-dependent:

$$\lambda_i = \frac{v_{d,i}}{H_{mix}} \tag{12}$$

where $v_{d,i}$ is the total deposition velocity (combining gravitational settling and surface impaction) and H_{mix} is the mixing height. Larger grains have higher deposition velocities, resulting in faster atmospheric removal:

$$v_{d,i} = v_{s,i} + v_{imp,i} \tag{13}$$

The impaction velocity v_{imp} depends on grain inertia (Stokes number) and surface roughness. For campus-scale transport over grass and pavement, typical values are $v_{imp} \approx 0.2$–0.5 cm/s for small grains and 1–2 cm/s for large grains.

3.5.4 Turbulent Diffusivity

The lateral (K_y) and vertical (K_z) eddy diffusivities in Eq. (3) are properties of the atmosphere, not the particle. However, the *effective* vertical diffusivity for heavy particles is reduced because gravitational settling counteracts upward turbulent mixing:

$$K_{z,eff,i} = K_z \cdot \exp\left(-\frac{v_{s,i}}{u_*}\right) \tag{14}$$

where $u_* \approx 0.1$–0.4 m/s is the friction velocity. For small Oak pollen ($v_s = 0.5$ cm/s $\ll u_*$), the correction is negligible ($K_{z,eff} \approx K_z$), and the particle behaves as a passive tracer. For heavy Pine pollen ($v_s = 3.1$ cm/s), the effective diffusivity is reduced by $\sim$10%, confining the plume closer to the ground.

3.5.5 Species-Specific Plume Behavior Summary

Combining these effects, the full species-dependent Gaussian plume is:

$$C_i(x', y', z) = \frac{Q_i(t)}{2\pi U \sigma_y \sigma_{z,i}} \exp\left(-\frac{y'^2}{2\sigma_y^2}\right) V_i(z, x') \cdot e^{-\lambda_i x'/U} \tag{15}$$

This produces qualitatively different spatial patterns by species:

- **Oak** (v_s small, λ small): Wide, far-reaching plumes. Concentration remains elevated hundreds of meters downwind. Dominates campus-wide background.
- **Pine** (v_s large, λ large): Narrow, rapidly decaying plumes. High concentration within 20–50 m of source, then sharp falloff. Creates localized "hot spots" near pine trees.
- **Redwood/Elm** (intermediate): Moderate plume extent, 50–150 m effective range.

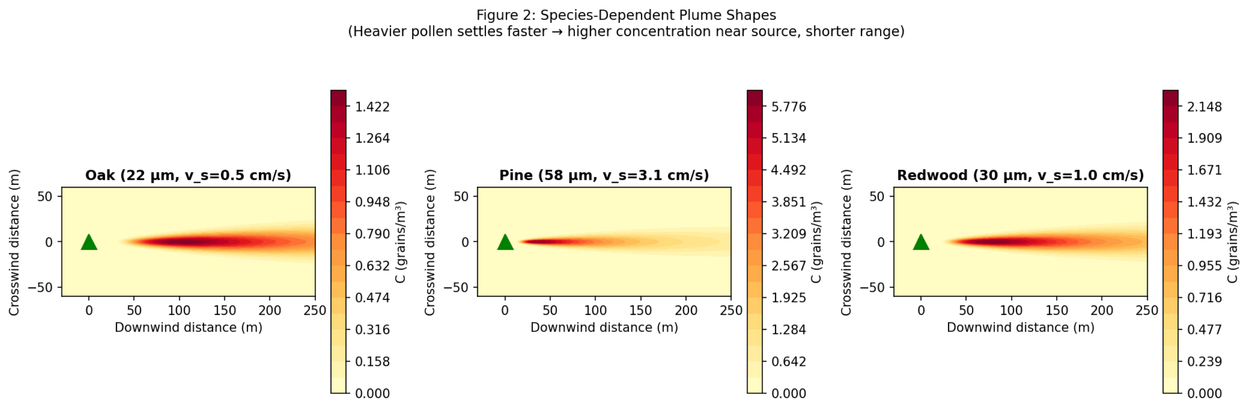

Figure 4: Species-dependent plume shapes due to different settling velocities. Oak pollen (small, light grains) produces wide, far-reaching plumes. Pine pollen (large, heavy grains) creates intense but localized concentration near the source tree.

3.6 Dispersion Coefficients

The crosswind (σ_y) and vertical (σ_z) standard deviations follow Pasquill-Gifford power-law relations [Gifford, 1961]:

$$\sigma_y = a_y (x')^{b_y}, \quad \sigma_z = a_z (x')^{b_z} \tag{16}$$

with coefficients (a, b) tabulated for atmospheric stability classes A–F (Table 2). Stability is estimated from wind speed and cloud cover using Turner's method [Turner, 1964].

Table 2: Pasquill-Gifford dispersion coefficients.

Class	a_y	b_y	a_z	b_z
A	0.22	0.894	0.20	0.894
B	0.16	0.894	0.12	0.894
C	0.11	0.894	0.08	0.894
D	0.08	0.894	0.06	0.894
E	0.06	0.894	0.03	0.894
F	0.04	0.894	0.016	0.894

3.7 Coupling to the Potential-Flow Wind Field

The potential-flow field of Section 2 supplies the wind; the Gaussian plume (Eq. (4)) supplies the dispersion. For each source i at location (x_i, y_i), we sample the local wind (u_i, v_i) from the converged field at that cell and convert it to a local speed $U_i = \sqrt{u_i^2 + v_i^2}$ and direction θ_i. The plume for source i is then evaluated with (U_i, θ_i) in place of a single campus-wide wind, so each plume bends and stretches according to the flow around its nearby buildings (Figs. 1–2). This is the mechanism by which the building-scale wind structure propagates into the pollen field: a tree in a channeled, high-speed gap produces a stretched, dilute plume, while a tree in a sheltered, low-speed pocket produces a compact, concentrated one. (For comparison, the system also retains a simpler uniform-wind mode with empirical building-downwash corrections, but the potential-flow coupling above is the default.)

3.8 Multi-Source Superposition

By linearity of Eq. (3), the total field from N sources is:

$$C_{total}(x, y) = \sum_{i=1}^{N} C_i(x, y) \tag{17}$$

evaluated via NumPy vectorized broadcasting over the full 120×120 spatial grid.

3.9 Building Footprint Masking

At $z = 1.5$ m within a building, there is no outdoor air exposure. Building rooftops detected from satellite imagery (Section 5.5) are rasterized to a binary mask, and the concentration inside them is zeroed:

$$C_{masked}(x, y) = C_{total}(x, y) \cdot (1 - M_{building}(x, y)) \,. \tag{18}$$

The mask is computed directly from the current building polygons, so it always matches the buildings displayed on the map.

4 Phenological Emission Model

4.1 Temporal Gate Function

Species-specific seasonal emission is modeled as:

$$Q_i(t) = Q_{base,i} \cdot W_i \cdot \Gamma_i(t) \tag{19}$$

The three factors are:

- $Q_{base,i}$ (**Base Emission Rate**, grains s^{-1}): The maximum pollen production rate of species i at peak bloom under ideal conditions. This represents the intrinsic reproductive output of the plant and depends on species physiology, tree maturity, and canopy volume. Values are derived from aerobiological literature; for example, a mature Coast Live Oak may release $\sim$500 grains/s at peak, while a Pine produces $\sim$400 grains/s. Formally:

$$Q_{base,i} = \rho_i \cdot A_{canopy,i} \tag{20}$$

  where ρ_i is the species-specific pollen flux density (grains $\mathrm{s}^{-1}\,\mathrm{m}^{-2}$) and $A_{canopy,i}$ is the canopy surface area. In our implementation, a per-species reference rate is assigned from literature values (Table 3) and then scaled by the mapped canopy size of each individual tree. Since a canopy's projected area scales with the square of its radius r_i, and each tree's radius is estimated during detection (and can be corrected by a reviewer), the per-tree base emission is

$$Q_{base,i} = Q_{ref,species(i)} \cdot \text{clip}\left(\left(\frac{r_i}{r_0}\right)^2,\ 0.1,\ 10\right), \tag{21}$$

where $r_0 = 4$ m is a nominal reference canopy radius and the clip keeps a hand-edited radius from producing physically implausible emission. Thus a large mapped oak emits proportionally more pollen than a small one of the same species.

- W_i (**Allergen Potency Weight**, dimensionless, 0–5): A clinical severity index quantifying the allergenic impact of species i relative to other pollen types. This accounts for the fact that equal grain counts of different species produce vastly different immune responses. For instance, Oak pollen ($W = 4.0$–4.5) triggers severe rhinitis and asthma at lower concentrations than Pine pollen ($W = 2.0$), whose large grains are less immunologically reactive. The potency weight scales the effective emission to reflect health impact rather than raw grain count:

$$Q_{effective,i}(t) = Q_{base,i} \cdot W_i \cdot \Gamma_i(t) \tag{22}$$

  Values are calibrated from clinical allergenicity databases and regional symptom prevalence data.

- $\Gamma_i(t)$ (**Temporal Gate Function**, dimensionless, 0–1): A smooth envelope controlling the seasonal on/off behavior of pollen release. Modeled as a truncated Gaussian:

$$\Gamma_i(t) = \begin{cases} \exp\left(-\frac{(t-t_{peak})^2}{2\sigma_t^2}\right) & t_{start} \le t \le t_{end} \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

Here t is the Julian day of year (1–365), t_{start} and t_{end} define the pollination window boundaries, t_{peak} is the day of maximum bloom intensity ($\Gamma = 1$), and σ_t (days) controls the width of the blooming bell curve. Outside the window $[t_{start}, t_{end}]$, the tree is dormant and releases zero pollen regardless of weather.

Physical Interpretation. The product $Q_{base} \cdot W \cdot \Gamma(t)$ yields a time-varying "effective allergenic emission" in potency-weighted grains per second. On the peak bloom day ($\Gamma = 1$) of a high-potency species ($W = 4.5$), the effective source strength is $Q_{base} \times 4.5$. By mid-summer when $\Gamma = 0$ for all tree species, the effective emission is zero regardless of Q_{base} or W.

4.2 Species Phenological Parameters

Table 4 lists the temporal parameters for the modeled genera, derived from Bay Area regional data [Bastl et al., 2016]. Most bloom in late winter through spring; Cedar (*Cedrus*) is a notable exception, pollinating in the fall (October–December) and producing a secondary autumn peak, while the campus-specific sets differ (e.g. Stanford adds Canary Island Palm, Olive, and Eucalyptus).

4.3 Generic "Other" Species Class

Reliable species-level identification from RGB satellite imagery alone is imperfect, and some campus trees belong to genera we do not model individually. We therefore assign each mapped canopy either to one of the modeled genera or to a generic **Other** class. To keep the labeling faithful to the real

Table 3: Base emission rates and potency weights by species.

Species	Q_{base} (grains/s)	W (potency)
Valley Oak	550	4.5
Coast Live Oak	500	4.0
Coast Redwood	300	2.5
Cedar	400	3.0
Pine	400	2.0
Chinese Elm	350	3.0
Western Sycamore	400	3.5
Canary Island Palm	350	3.0
Olive	600	5.0
Eucalyptus	250	2.0

Table 4: Species phenological parameters.

Species	t_s	t_e	t_p	σ_t	W
Valley Oak	60	120	95	21	4.5
Coast Live Oak	60	120	91	21	4.0
Redwood	32	91	60	21	2.5
Cedar	274	335	305	21	3.0
Pine	91	181	135	30	2.0
Chinese Elm	244	305	274	21	3.0
Sycamore	74	121	100	18	3.5
Canary Island Palm	74	135	105	20	3.0
Olive	105	150	128	18	5.0
Eucalyptus	1	120	60	30	2.0

campus rather than to imagery color alone, the Gunn species proportions are anchored to the 2009 PAUSD arborist inventory (Section 7): oaks and redwood dominate ($\sim$41% and $\sim$20% of canopies respectively), with smaller shares of cedar, sycamore, elm, and pine, and the remaining $\sim$22% (the inventory's long tail of ornamental/non-native genera) assigned to *Other*.

An *Other* tree still emits pollen: rather than being dropped from the dispersion sum, it is assigned a generic moderate-emitter profile—a base rate between that of a strong and a weak emitter, a mid-range potency, and a broad spring pollination window—so that unclassified canopies contribute a plausible background load without overstating the allergenic impact of any particular genus. The generic profile is deliberately conservative; refining an *Other* tree to a specific genus (Section 5.6) sharpens both its emission timing and its potency weighting. This design keeps the model honest about classification uncertainty: campus-wide background pollen is driven by the abundant oaks and redwoods, while the *Other* sources contribute a diffuse baseline.

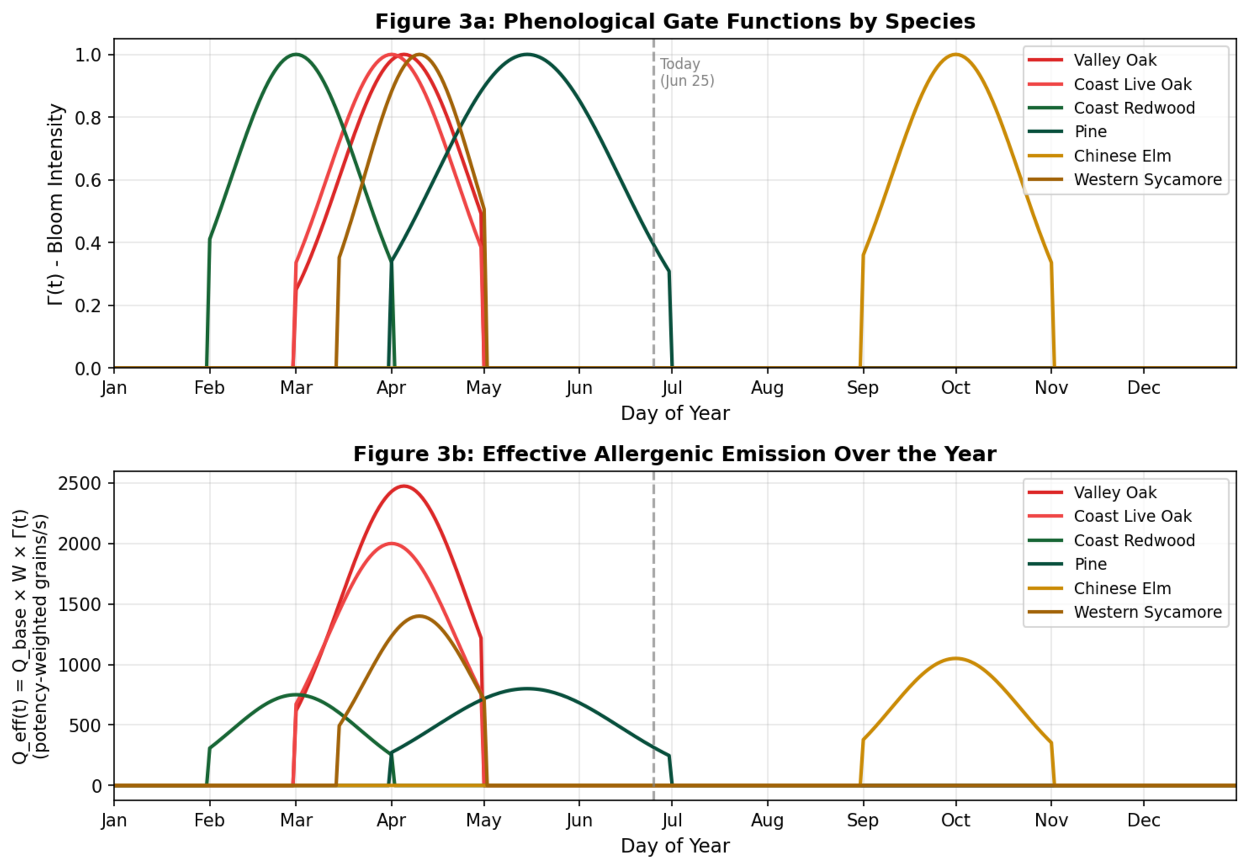

Figure 5: Phenological emission model. (a) Temporal gate functions $\Gamma_i(t)$ showing blooming intensity across the year for each species. (b) Effective allergenic emission $Q_{eff}(t) = Q_{base} \times W \times \Gamma(t)$. The dashed line marks today (June 25), when most tree species are dormant and only Pine retains weak late-season emission.

5 Satellite-Based Vegetation Detection

5.1 Image Acquisition

High-resolution satellite imagery (0.5 m/pixel) is obtained from the Esri World Imagery REST export API, covering the full campus (557 m × 637 m) at 1600 × 1398 pixels.

5.2 Calibrated Color Segmentation

Tree canopy pixels are classified using thresholds calibrated from 771 human-verified positive samples:

$$\text{is\_tree} = (g_e > 4) \wedge (35 < \bar{I} < 155) \wedge (r < 165) \wedge (b < 125) \tag{24}$$

where $g_e = g - (r+b)/2$ is the green excess and $\bar{I} = (r+g+b)/3$ the mean brightness.

5.3 Watershed Segmentation

Adjacent canopies are separated using marker-controlled watershed [Beucher and Meyer, 1992]:

1. Binary mask is eroded (3 iterations) to generate seed markers

2. Euclidean distance transform computed on the mask
3. Watershed algorithm segments using inverted distance as elevation

5.4 Peak-Based Detection

An alternative pipeline computes a "tree-ness" score:

$$S(x, y) = \max(g_e, 0) \cdot \frac{\max(155 - \bar{I}, 0)}{120} \tag{25}$$

convolved with a Gaussian ($\sigma = 4$ px). Local maxima with minimum spacing of 7 pixels identify individual tree centers. Canopy radius is estimated from the score falloff profile.

5.5 Building Rooftop Detection with Deep-Learning Segmentation

Rooftops are harder to isolate by color alone than tree canopies, because roofs, pavement, and dry ground occupy overlapping color ranges. We therefore detect buildings with the Segment Anything Model (SAM) [Kirillov et al., 2023], a pretrained promptable segmentation network. SAM's automatic mask generator proposes class-agnostic masks over the campus image; we then retain only masks that look like rooftops, using a color test (gray/white or warm red-tile roofs, with vegetation excluded), a building-scale size test, and a compactness test. Because SAM downsamples its input internally, we run it on a grid of overlapping image tiles so that small rooftops are resolved at higher effective resolution, then merge detections with spatial de-duplication. Each retained mask is reduced to a rotated minimum-area rectangle, so angled and non-axis-aligned buildings are represented at their true orientation. SAM inference is run once at cache-generation time and the results are stored; the deployed server never runs the network.

5.6 Human-in-the-Loop Calibration

Both tree and building detections undergo iterative refinement through an interactive map editor: automated detection proposes candidates, and a reviewer corrects them directly on the satellite basemap. For **trees**, the reviewer can delete false positives (e.g. shrubs or shadows mistaken for canopy), relocate a canopy by dragging it to its true position, resize its canopy radius (which rescales that tree's emission via the canopy-area law of Eq. (21)), and reclassify its species—including promoting a generic *Other* tree (Section 4.3) to a specific genus or vice versa. For **buildings**, the reviewer edits each footprint's name, floor count, position, and shape; footprints are not constrained to axis-aligned rectangles but may be dragged into arbitrary convex quadrilaterals (parallelograms, trapezoids) to match obliquely oriented or non-rectangular rooftops, and rotated freely. Each correction is persisted to the per-campus detection cache that the server serves, so the physical model (emission strengths, indoor exclusion masks, and the potential-flow obstacle field) always reflects the reviewed geometry. For each building the height is taken as (floors $\times$ 3 m), which sets the indoor exclusion volume.

5.7 Multi-Campus Imagery

The pipeline is campus-agnostic: a campus is defined by its geographic bounds, from which the satellite image and detections are generated. Gunn High School uses Esri World Imagery; the Stanford core uses USGS NAIP imagery (whose true-color rendering shows the characteristic red-tile roofs), fetched seam-free by stitching Web-Mercator tiles.

5.8 Rooftop-Overlap Filtering

Color-based canopy detection occasionally fires on rooftops—warm or textured roof pixels can resemble vegetation—especially on dense urban campuses. After buildings are detected, we therefore prune tree candidates geometrically: a canopy is discarded if more than 50% of its disk area (estimated by area-weighted sampling of the disk against the building polygons) lies inside any detected footprint. This is applied only where needed; on the densely built Stanford core it removed a substantial share of the raw canopies (Section 7), leaving trees in courtyards, along streets, and on open ground while eliminating rooftop false positives.

6 Exposure Assessment

6.1 Path-Integrated Dose

Cumulative exposure along walking path γ at speed $v_w = 1.4$ m/s:

$$D = \int_\gamma C_{total}(\mathbf{x}(s)) \cdot \frac{ds}{v_w} \approx \sum_k C(\mathbf{x}_k) \cdot \Delta t_k \tag{26}$$

6.2 Clinical Risk Mapping

Dose is mapped to actionable risk levels: Low ($D < 50$), Moderate (50–200), High (200–500), Very High (> 500 grain·s/m^3).

7 Results

7.1 Detection Performance

On the Gunn campus the detection pipeline maps roughly 900 trees, which we label against the seven allergenic genera modeled here (Valley Oak, Coast Live Oak, Coast Redwood, Cedar, Western Sycamore, Pine, and Chinese Elm) plus a generic *Other* class (Section 4.3); the exact counts evolve as the set is refined through the interactive editor. The species proportions are anchored to the 2009 PAUSD arborist inventory of the campus [PAUSD, 2009], which surveyed 455 numbered trees. In that inventory the dominant genera are Valley Oak (116 trees, 25.5%), Coast Redwood (92, 20.2%), Coast Live Oak (70, 15.4%), Cedar (27, 5.9%), Western Sycamore (21, 4.6%), and Chinese Elm (20, 4.4%); all oaks together account for 192 trees (42.2%). The remaining ∼17% comprises a long tail of ornamental and non-native genera (e.g. hawthorn, privet, sweetgum, pepper) that are not individually modeled and are assigned to the *Other* class. On the larger Stanford core, tiled SAM detection identifies on the order of 10^3 building rooftops and $\sim 10^4$ tree canopies, giving comprehensive coverage of the full campus. Because tree detection on the dense Stanford imagery placed many canopy candidates on top of rooftops, we apply a geometric filter after building detection: a candidate tree is discarded when more than half of its canopy disk area falls inside any detected building footprint (evaluated by area-weighted sampling of the disk against the building polygons). This removed roughly 43% of the raw Stanford canopies (from ∼9,900 to ∼5,650), eliminating rooftop false positives while retaining trees in courtyards, streets, and open ground. The surviving Stanford canopies are labeled against the six most abundant genera reported for the campus [Trees of Stanford, 2026, Stanford LBRE, 2026]—Coast Live Oak (the single most numerous species, ∼40% of the population), Canary Island Palm, Eucalyptus, Coast Redwood, Valley Oak, and Olive—with all remaining genera assigned to *Other*. Detection and imagery are cached per campus, so no machine-learning model runs on the live server.

7.2 Seasonal Concentration

Using a fixed scale (500 grains/m^3 = spring peak):

- **April 1** (day 91): $C_{max} = 273$ grains/m^3 (55% of scale); 8 active species
- **June 25** (day 176): $C_{max} = 30$ grains/m^3 (6%); Pine only
- **January** (day 15): $C_{max} \approx 0$; all species dormant

7.3 Spatial Patterns

The concentration field exhibits: (i) plumes bending along the potential-flow streamlines as the wind is diverted around buildings; (ii) zero concentration over building footprints; (iii) locally elevated concentration where flow is channeled between closely spaced buildings; and (iv) elevated concentrations in oak-dense corridors.

8 Discussion

The coupled potential-flow and Gaussian-plume approach is computationally efficient (a few seconds per full field, even with thousands of sources) and suitable for real-time interactive use. It captures how wind is diverted and channeled around buildings, which a spatially uniform wind cannot. Its key physical limitation is that potential flow is inviscid: it produces no turbulent wake or recirculation behind buildings, so pollen trapping in lee zones is not represented; a viscous CFD treatment would be required for that. Other limitations include RGB-only tree-species classification (improvable with hyperspectral data), uniform assumed building heights where floor counts are unknown, and the absence of ground-truth pollen measurements for quantitative validation.

The human-in-the-loop calibration workflow drove most of the accuracy gain: automated detection alone achieved approximately 75% precision, improved to $> 95\%$ after three interactive correction rounds. For campus-scale applications, then, semi-automated approaches combining algorithmic detection with targeted human verification are more practical than fully autonomous methods.

The fixed annual scale reveals that allergy risk communication should emphasize seasonal context: a June reading of 30 grains/m^3 might alarm students if presented without reference to the 500 grains/m^3 spring peak that constitutes the actual high-risk period.

8.1 Future Work: Coupling Humidity, Temperature, and Precipitation

The current model is driven by wind speed, wind direction, and atmospheric stability class. Relative humidity and temperature are retrieved from the weather service and shown to the user, but they do not yet modify the dispersion calculation (temperature only informs the stability-class estimate). A natural and high-value extension is to couple humidity, temperature, and precipitation into emission and removal, and we outline the intended formulation here as future work rather than as an implemented result.

First, most anemophilous species suppress pollen release under high humidity because anthers must dry to dehisce; this could scale emission by a humidity factor,

$$Q_i(t, RH) = Q_i(t)\, f_{RH}(RH), \qquad f_{RH}(RH) = \begin{cases} 1.0 & RH < 50\% \\ 1.0 - 0.8\,\dfrac{RH - 50}{50} & 50\% \leq RH < 90\% \\ 0.1 & RH \geq 90\%, \end{cases} \tag{27}$$

so that emission falls to $\sim$10% of its dry-air value in foggy, marine-layer conditions. Second, hygroscopic grains swell in humid air, increasing their settling velocity, $d_p(RH) = d_{p,dry}\,[1 + \kappa\, RH/(100 - RH)]^{1/3}$ with $v_s \propto d_p^2$, which would enhance near-source deposition. Third, rainfall scavenges airborne pollen, adding a wet-removal term $\lambda_{wet} = \Lambda P^{0.7}$ (with P the rainfall rate) to the first-order decay. Finally, rapid humidity spikes above $\sim$95% can rupture grains into respirable sub-pollen particles implicated in "thunderstorm asthma," which could be modeled as a conditional secondary source with near-zero settling velocity. Figure 6 illustrates the qualitative form of the emission-suppression and hygroscopic-swelling responses. Implementing and, critically, validating these couplings against measured pollen and weather time series is a priority for future work.

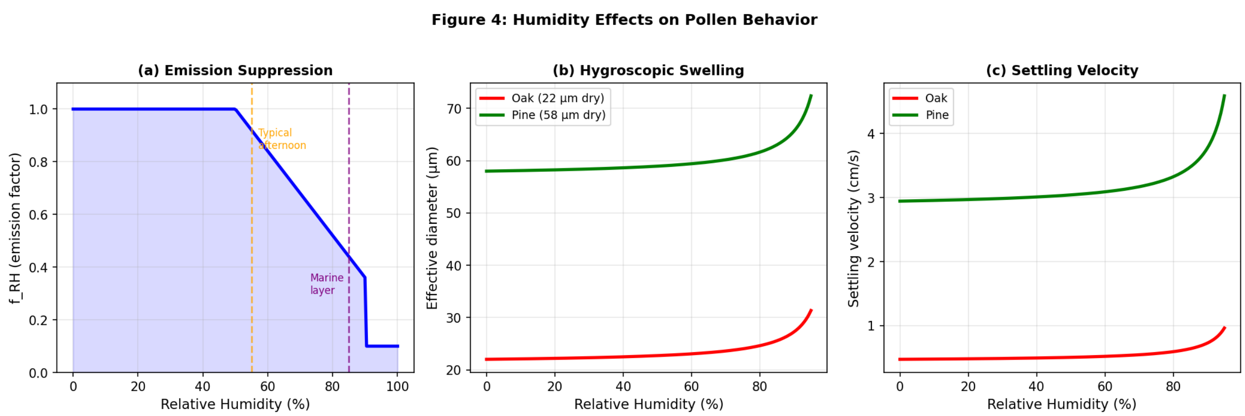

Figure 6: Proposed humidity coupling (future work; not part of the current model). (a) Emission suppression factor f_{RH}: pollen release drops sharply above 50% RH. (b) Hygroscopic swelling increases grain diameter at high humidity. (c) Resulting increase in gravitational settling velocity.

9 Conclusion

We built a complete pipeline for hyperlocal pollen dispersion modeling, from satellite-based source identification, through a potential-flow wind field coupled to Gaussian-plume transport, to path-integrated exposure. The system resolves concentration gradients at meter scale across two campuses of contrasting scale—Gunn High School and the Stanford core—with thousands of individually mapped trees and buildings, and recomputes the full field in a few seconds as wind and season change. Whether the routes it recommends actually reduce measured exposure is the validation step that remains.

Data Availability

The complete source code and campus data are publicly available at `https://github.com/jasminejz2021ai/pollenWebApp`. A live, interactive deployment of the system is accessible at `https://pollen-web-app-seven.`

`vercel.app/`. The tree inventory is derived from the PAUSD 2009 arborist survey supplemented by satellite detection.

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